Darshan Sapnar

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Education

K.K. Wagh Institute of Engineering Education and Research Aug. 2023 – June 2027
Bachelor of Technology (B.Tech) in Information Technology *CGPA: 7.93/10.00* *Nashik, Maharashtra*

Experience

CareerGyan – Career Guidance Platform Apr. 2026 – June 2026
Freelance Developer | Laravel, MySQL, Blade, OpenAI API, GitHub Actions GitHub

- Integrated an **OpenAI GPT-4o Mini** chatbot with session-authenticated access and backend **rate-limiting middleware** to prevent API abuse, synchronizing quota state asynchronously on the frontend.
- Set up a **GitHub Actions** CI/CD pipeline for automated deployment to MilesWeb shared hosting.
- Designed and built responsive frontend pages for the platform's career exploration and chatbot interfaces.

Projects

DAuth – OpenID Connect Identity Provider May 2026 – June 2026
React, Node.js, Express.js, PostgreSQL, Prisma ORM, JWT, OIDC, OAuth 2.0 GitHub

- Built a self-hosted identity provider extending **OAuth 2.0** authorization with **OIDC** authentication, handling user login, sessions, OAuth client management, and signed **JWT** issuance.
- Implemented **Authorization Code Flow with PKCE**, refresh tokens, client registration, and redirect URI validation to secure the authentication flow.
- Designed a React admin dashboard to manage users, OAuth clients, active sessions, and app settings.

Relay – AI-Powered Video Conferencing Platform Aug. 2026 – Sept. 2026
React.js, Node.js, Express.js, LiveKit, WebRTC, Socket.IO, PostgreSQL (Neon) GitHub

- Built a real-time video conferencing platform supporting multi-participant audio/video communication and screen sharing using **LiveKit** and **WebRTC**.
- Implemented real-time group chat and direct messaging with **Socket.IO**, persisted in **PostgreSQL** (Neon).
- Integrated an **AI meeting assistant** for live transcription, summaries, key decisions, and action items via LLMs.
- Built meeting access controls with waiting rooms, host permissions, participant removal, and meeting locking.

LLM Gateway – Fault-Tolerant AI Proxy with Caching Sept. 2026 – Oct. 2026
Node.js, Express.js, React, MongoDB, Redis Stack, Docker, GitHub Actions, Groq, Gemini GitHub

- Engineered a multi-provider LLM gateway with automatic **failover**, retries, circuit breaking, and **SSE streaming**, with an interactive Playground and monitoring dashboard; automated CI/CD using GitHub Actions with **71 tests**.
- Optimized LLM response latency using exact and semantic caching with Redis HNSW vector search, achieving **2–4 ms cache-hit latency** and a **41% cache hit rate** across 32 test requests, including **25% via semantic matching**, with an estimated **38% reduction in API costs**.
- Designed a **verification layer** (antonym guard, entity check, LLM judge) for semantic matches, raising the cache hit **to 100% with 0 wrong answers** on a held-out test set.
- Implemented **PII redaction** across **5 data types**, correctly masking every sensitive value with **zero false alarms** across **40 test cases**, alongside **token-based rate limiting** and **API cost tracking**.

Technical Skills

Languages: C, C++, Python, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks: React.js, Node.js, Express.js, Tailwind CSS, Laravel, Rest APIs, LLM APIs
Developer Tools: Git, GitHub, GitHub Actions, VS Code, Docker, Postman, Jupyter Notebook, Chrome DevTools
Databases & Cloud: MySQL, MongoDB, PostgreSQL, Prisma ORM
Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), Problem Solving, Operating Systems, Computer Networks, Caching, Vector Search, CI/CD

Certification

Full Stack Web Development Cohort – ChaiCode *(MERN, REST APIs, authentication, AI integration)* July 2026